

NORTHEAST ALABAMA, AL, USA

WMO: 722285

Lat: 33.967N

Lon: 86.083W

Elev: 569

StdP: 14.40

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	22.7	5.3	7.4	23.9	9.8	9.3	28.5	19.2	44.6	17.4	44.5	3.5	340	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	94.2	75.0	91.6	74.9	90.3	74.7	78.9	88.1	77.8	87.4	76.8	86.2	6.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	141.1	83.1	75.1	134.5	82.3	73.5	127.4	81.1	42.8	88.2	41.6	87.3	40.5	87.2	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	14.2	12.2	12.5	96.4	4.9	3.7	9.0	99.1	6.1	101.3	3.4	103.3	-0.1	106.1
			11.3	80.1	4.3	1.5	8.2	81.2	5.7	82.1	3.3	82.9	0.2	84.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	41.6	46.6	53.5	60.4	68.7	75.9	79.1	78.5	72.7	61.6	50.4	43.8
	DBStd	15.11	9.98	9.29	8.82	7.21	6.07	3.97	3.27	3.61	5.95	8.02	8.19	8.92
	HDD50	875	301	158	68	9	0	0	0	0	0	9	93	237
	HDD65	3086	726	516	366	171	37	1	0	0	12	159	441	657
	CDD50	4945	40	64	175	321	581	777	902	884	680	369	106	46
	CDD65	1682	1	2	9	33	153	328	437	419	242	54	3	1
	CDH74	15390	1	13	120	435	1486	2945	3992	3764	2074	526	33	1
	CDH80	6306	0	0	13	64	472	1263	1819	1736	811	124	4	0
(14) Wind	WSAvg	4.3	5.3	5.3	5.6	5.0	4.0	3.5	3.3	3.2	3.8	4.0	4.2	4.8
(15) Precipitation	PrecAvg	55.2	5.2	5.3	5.8	4.8	4.2	4.5	5.1	4.0	3.8	2.9	4.3	5.2
	PrecMax	74.1	8.7	9.8	9.3	9.0	13.3	10.7	9.7	8.7	8.8	9.8	10.5	9.8
	PrecMin	29.3	2.3	2.5	1.8	1.5	0.6	2.1	1.3	0.6	0.6	0.1	1.3	2.1
	PrecStd	9.1	1.9	1.8	1.8	1.9	2.6	2.2	2.2	1.7	2.4	2.2	2.0	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	75.5	81.6	84.4	90.5	95.1	97.2	98.9	94.7	87.9	79.1	72.1
		MCWB	60.8	62.8	64.2	66.1	72.9	74.1	75.8	75.3	73.0	71.4	65.1	61.9
	2%	DB	66.3	71.0	77.3	81.8	88.0	91.5	93.4	94.2	91.2	82.9	73.0	66.8
		MCWB	58.8	61.3	62.4	65.4	71.8	73.8	75.8	75.5	72.9	68.6	61.6	60.5
	5%	DB	63.0	66.5	73.4	79.4	84.5	90.1	91.2	91.1	88.2	80.5	70.1	63.6
		MCWB	56.5	58.6	59.9	63.8	70.3	74.1	76.0	75.2	72.3	65.7	60.1	57.5
	10%	DB	58.7	63.3	70.2	75.5	82.2	87.8	89.6	89.6	84.3	75.7	65.9	59.3
		MCWB	53.0	54.8	58.4	62.3	69.2	73.6	75.9	74.9	71.5	64.8	58.1	53.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.0	67.3	67.3	70.4	76.2	79.8	80.5	80.0	77.7	75.6	68.7	65.7
		MCDB	68.2	71.6	75.7	79.8	84.3	87.3	88.6	89.7	86.4	82.5	72.6	69.4
	2%	WB	60.6	64.2	65.2	68.2	74.3	77.6	79.2	78.4	76.1	72.4	65.6	61.8
		MCDB	64.7	68.6	73.4	77.6	83.4	87.2	88.2	88.5	85.0	78.9	71.5	65.6
	5%	WB	57.9	60.5	62.9	66.4	72.6	76.3	78.0	77.5	74.7	68.9	62.4	58.6
		MCDB	62.1	64.9	70.4	75.8	81.9	85.6	87.5	87.6	83.6	76.0	67.4	63.3
	10%	WB	53.6	56.8	60.4	64.5	71.1	75.0	76.9	76.3	73.2	66.6	58.9	54.6
		MCDB	57.7	61.2	67.9	72.9	79.9	83.7	86.4	85.7	81.7	74.1	64.6	58.7
(35) Mean Daily Temperature Range	5% DB	MDBR	20.2	21.9	23.6	25.4	23.3	21.0	19.6	20.6	21.9	24.4	24.6	20.5
		MCDBR	24.4	26.9	28.8	28.9	27.1	25.7	22.7	24.6	25.7	28.2	28.9	24.7
	5% WB	MCWBR	18.0	18.0	16.5	15.1	12.9	10.8	9.0	9.5	10.4	14.6	18.6	18.7
		MCWBR	19.7	21.4	23.4	23.7	22.4	21.0	20.4	20.9	20.7	21.3	22.0	20.2
(40) Clear-Sky Solar Irradiance	taub		0.320	0.323	0.363	0.395	0.461	0.486	0.535	0.536	0.434	0.376	0.343	0.322
		taud	2.506	2.493	2.404	2.299	2.156	2.127	2.007	1.992	2.271	2.418	2.450	2.515
	Ebn at Noon		283	294	289	283	264	257	243	240	263	271	271	276
		Edh at Noon	27	30	36	41	48	49	56	56	40	32	28	25
(44) All-Sky Solar Radiation	RadAvg		816	981	1342	1731	1892	1955	1876	1739	1512	1255	922	694
	RadStd		58	126	123	117	152	152	121	95	138	162	100	70

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	+284	+203

Nomenclature: See separate page